TRANSMISSION—5R55N—DESCRIPTION OF FEATURES AND MAJOR COMPONENTS	Article No. 99-8-13

LINCOLN: 2000 LS

ISSUE

Beginning with Job 1, April 19, 1999 production, the 2000 Lincoln LS will have a new electronically controlled 5-speed automatic transmission, the 5R55N.

ACTION

Refer to the following information and figures for a general description of 5R55N transmission features, major components, and available service and training information.

The 5R55N is our new 5-speed (Figures 1 and 2), RWD, non-synchronous transmission with 745 N•m (550 lb-ft), 6500 rpm input capability. Its applications include the new Lincoln LS and Jaguar X200. The internal structure of this transmission is very similar to 5R55E, which is the parent transmission. However, the 5R55N shares very few common components with 5R55E due to major upgrades throughout the development process.

The transmission features are as follows:

Electro-Hydraulic Shift Control Features

- Torque Converter Clutch Modulation for smooth clutch engagement and shift
- Full electronic shift control by our new 32-bit Powertrain Electronic Controller (PTEC) for smooth and responsive powertrain performance
  - Closed Loop Shift Control, Electronic Clutch Pressure Control and Engine Torque Modulation for smooth shift
  - Adaptive Shift Control for consistent shift quality throughout the life of the vehicle
  - Reactive Shift Scheduling to react to throttle position and the rate of throttle change
- Overdrive (5th gear) lockout in automatic mode in D4 shifter position (Figure 3)
- Manual Shift Capability (optional Select Shift Transmission (SST) (Figure 4)) to allow the driver to control the transmission shift point by tapping the shift lever forward or rearward
- Three (3) speed sensors for precise shift control (Transmission Speed Sensor (TSS), Idle Speed Sensor (ISS), and Output Shaft Speed Sensor (OSS))
- Integrated Solenoid Body Assembly (Figure 5), contains: three (3) Pressure Control (PC A/B/C) Solenoids, four (4) Shift Solenoids (SS A/B/C/D), Torque Converter Clutch (TCC) Solenoid, and a Transmission Fluid Temperature (TFT) Sensor
- Reverse Pressure Switch (Figure 6) to detect pressure in the main control valve body (mounted on the main control).

Hydraulic/Mechanical Features

- Wide ratio span 5-speed with close ratio steps for smooth shift quality and improved performance

Gear	1st	2nd	3rd	4th	5th	Reverse	Ratio Span
Ratio	3.25	2.44	1.55	1	0.75	3.07	4.33

- Three (3) One-Way Clutches for smooth non-synchronous shift
- Engine braking capability during coast operation in all gears
- One-piece die cast aluminum case with the re-enforcement ribs to minimize powertrain Noise/Vibration/Harshness (NVH)
- Fixed output flange to minimize driveline disturbances
- High capacity torque converter clutch to maximize the lockup capability in all throttle condition
- Brazed torque converter blades for high speed capability and improved efficiency
- Dedicated air-to-oil transmission cooler for low transmission operating temperature
- High output trochocentric pump with flow control capability
- Friction on steel plates (single sided clutch plates)
- New design Reverse Servo Piston Assembly
- Semi-synthetic fluid for improved durability - MERCON® V for service
- Transmission fluid leveling pipe which eliminates the need of conventional fill tube and dipstick - this new feature requires a special fluid fill procedure as described in the transmission Workshop Manual

- Adapter plate between the torque converter and flywheel assembly (V8 application only)
- Single bolt transmission rear mount
- Heated Exhaust Gas Oxygen (HEGO) sensors attached to the transmission oil pan
- Transmission wire harness with five (5) connectors.

MODEL COMPLEXITY				
Transmission Assembly (-7000-)	**Model Code (ID Tag Color)**	**Vehicle Application**	**Engine**	**Axle Ratio**
XR8P-AC	RJL-D (Apricot)	Jaguar X200	4.0L, V8	3.31
XR8P-BC	RJL-C (Light Green)	Jaguar X200	3.0L, V6	3.31
XW4P-AC	RJL-B (Yellow)	Lincoln LS	3.9L, V8	3.58
XW4P-BC	RJL-A (White)	Lincoln LS	3.0L, V6	3.58

Diagnostics
Diagnosis of the 5R55N will be similar to the 5R55E, although there are some unique features and function of the Output State Control (OSC) Mode not found in the 5R55E. Use NGS Tester or the new Worldwide Diagnostic System (WDS) and OSC function for diagnosis of the 5R55N transmission concerns. Follow the OSC procedures as listed in the 2000 LS Workshop Manual. Never short cut the diagnostic procedures or incorrect results may occur.

NEW SERVICE TOOLS/PROCEDURES		
Service Tool Description	**Service Tool Number**	**Used In Procedure(s)**
Output Shaft Flange Remover	307-408	Removal of the output shaft flange
Output Shaft Flange Installer	307-404	Installation of the output shaft flange (NOTE: This tool is used with Tool 205-479)
Air Test Plate/Gasket	307-405	Air checking of the transmission
Input Shaft Seal Remover	308-375	Removal of the input shaft seal
Transmission Pump Puller	307-397	Removal of the fluid pump from the case
Cushion Spring Compressor	307-401	Compressing the retaining ring for the removal of the intermediate clutch - also used to provide a pre-load when determining the #4 Thrust Bearing and #1 Thrust Washer
Servo Cover Compressor	307-402	Removal of the front and intermediate servo covers
Gauge Bar	307-400	Determining the #4 Thrust Bearing and #1 Thrust Washer
Flexplate Alignment Tool	307-403	Properly aligning the flexplate adapter plate, on V8 model, to the torque converter, preventing NVH issues
Alignment Pins, Bump Body (2 req'd)	307-398	Used to properly align pump halves during assembly
Alignment Pins, Pump To Case (2 req'd)	307-399	Used to properly align pump assembly into case during transmission assembly

Training and Service Information
The following is a list of training course and service information that is available to aid the service technician in understanding, diagnosing, and servicing the 5R55N transmission.

Training

- 2000 Lincoln LS New Model Training - Course Code Number: 34N18FO; FCS Number: FCS-12921-DL.

Service

- 2000 Lincoln LS Workshop Manual - Section 307-01, Automatic Transmission
- 2000 Lincoln LS Workshop Manual - Section 307-02, Automatic Transmission - Cooling
- 2000 Lincoln LS Workshop Manual - Section 307-05, Automatic Transmission - External Controls
- 2000 Lincoln LS Wiring Manual

Other Helpful Training Available

- STST Electrical Course No. 3000-214
- Automatic Transmission Electronic Control Systems Course No. CTP-1993-1
- Automatic Transmission Hydraulic/Mechanical Operation Course No. CTP-1992-3

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 504000

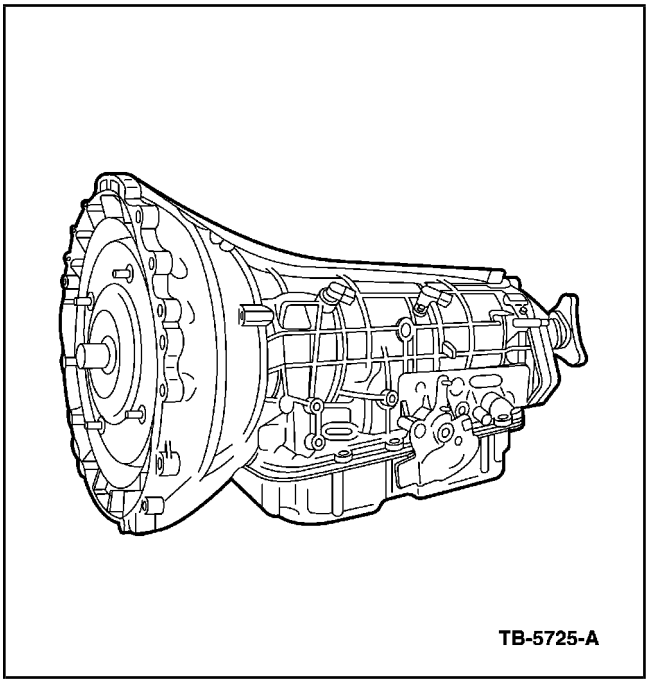

Figure 1 - Article 99-8-13

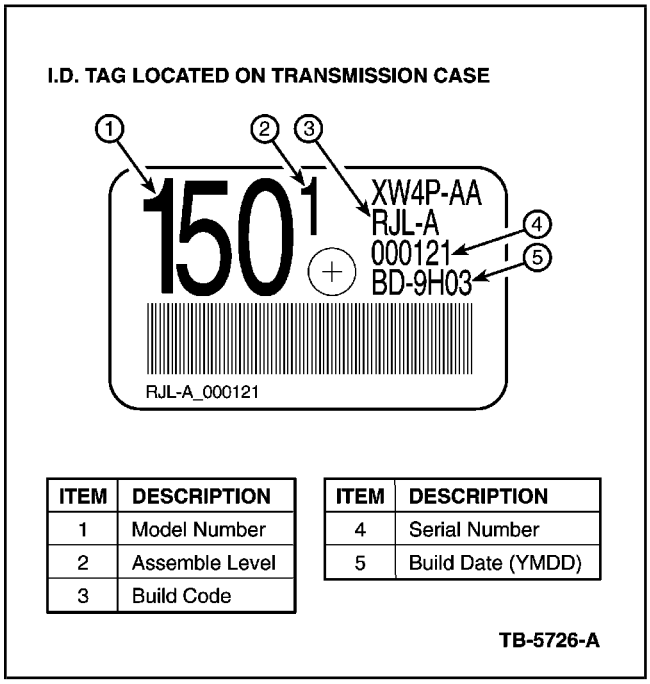

Figure 2 - Article 99-8-13

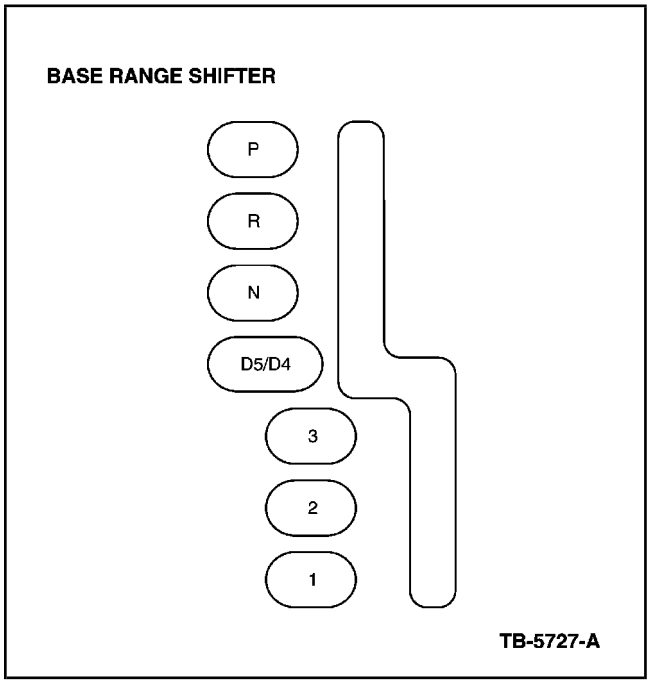

Figure 3 - Article 99-8-13

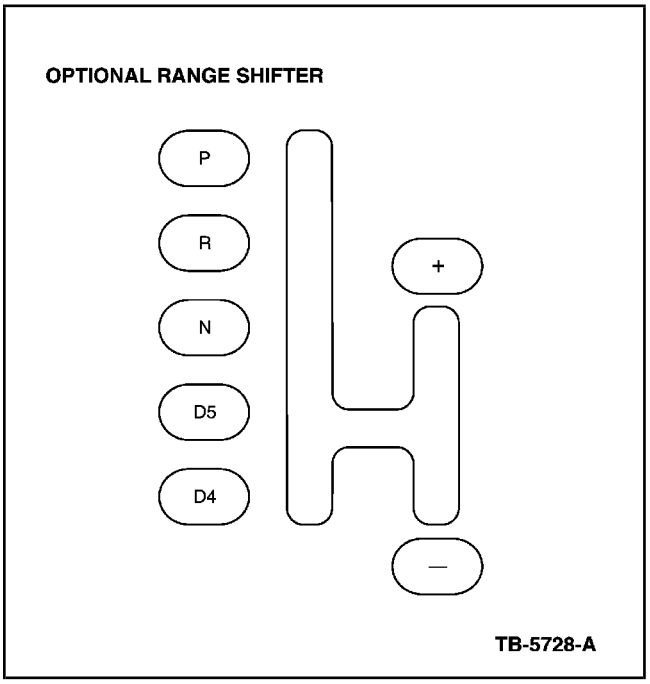

Figure 4 - Article 99-8-13

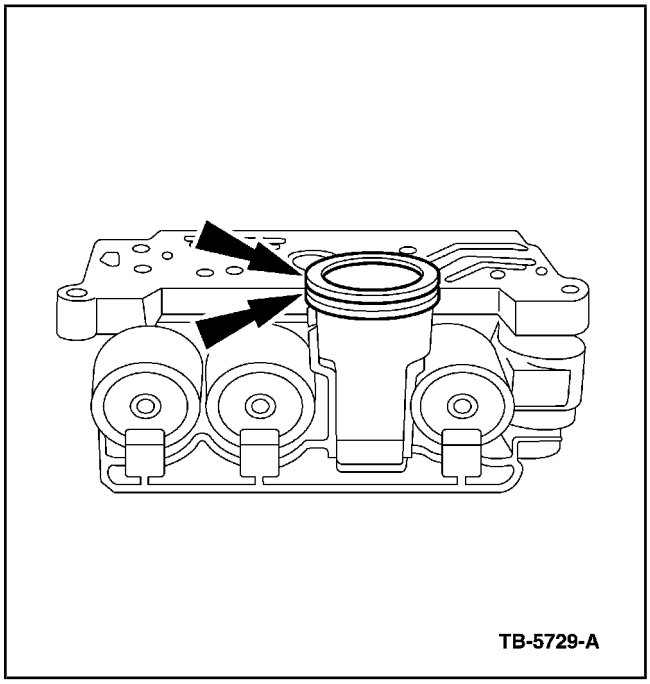

Figure 5 - Article 99-8-13

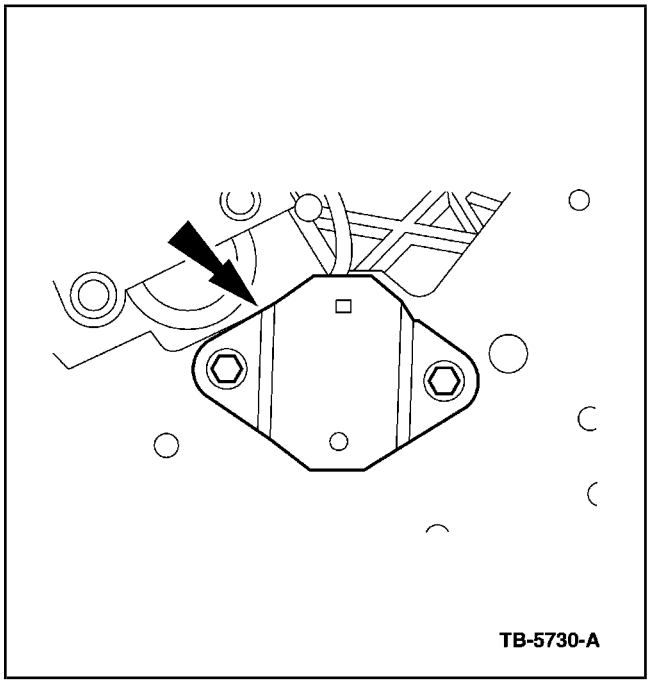

Figure 6 - Article 99-8-13